■ SalesRadar

Complete Study Guide

Technologies • Architecture • How It Works

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8 Modules	10 REST APIs	2 Languages	6+ Libraries
Full-Stack	AI Powered	ML Forecasting	PDF Reports

PART 1: PROJECT OVERVIEW

SalesRadar kya hai? (Hinglish)

SalesRadar ek **AI-powered full-stack analytics platform** hai jo car dealerships ke liye banaya gaya hai. Isme **Java (Spring Boot)** backend handle karta hai aur **Python (Streamlit)** frontend/ML karta hai. Dono technologies ko ek saath use karke ek powerful dashboard banaya jisme CSV upload, real-time charts, AI questions, ML prediction, aur PDF generation sab kuch hai.

English (Interviewer ko batao):

"SalesRadar is a full-stack automotive sales analytics platform that bridges Java's enterprise capabilities with Python's data science ecosystem. Car dealerships can upload sales CSV data and get instant insights through Plotly visualizations, AI-powered natural language querying via Spring AI + Ollama, ML-based forecasting using scikit-learn, and automated PDF report generation using iTextPDF."

8 Key Features:

Module	Description
■ CSV Upload	Auto clears old data, fresh analysis every time
■ KPI Summary	9 business metrics in one glance
■ Yearly Analytics	Year-wise trends with line & bar charts
■ Monthly Analytics	Month-wise breakdown for any year
■ Deep Analytics	Brand / Color / State wise charts
■ ML Prediction	scikit-learn Linear Regression — 1-5 year forecast
■ AI Insights	Natural language querying via Spring AI + Ollama
■ PDF Reports	Auto-generated reports via iTextPDF

PART 2: ARCHITECTURE — KAISE KAAM KARTA HAI?

Project 2 main services mein divided hai — Spring Boot (Java) port 8080 pe aur Streamlit (Python) port 8501 pe. Dono HTTP REST API calls se communicate karte hain.

■ USER (Browser — localhost:8501)

■ HTTP Requests (requests library)

■ STREAMLIT FRONTEND (Python)
Plotly Charts | scikit-learn ML | File Upload | PDF Download

■ REST API Calls (JSON)

■ SPRING BOOT BACKEND (Java — Port 8080)
Controller → Service → Repository (Layered Architecture)

■ JPA / Hibernate

■ MYSQL DATABASE
car_sales table | Aggregation Queries

Spring Boot ke saath ye services bhi connected hain:

Service	Technology	Kaam
AI Queries	Spring AI + Ollama	Natural language questions ka jawab
PDF Generation	iTextPDF 5	Automatic PDF reports banana
ML Prediction	scikit-learn (Python)	Future sales forecast karna
CSV Parsing	Apache Commons CSV	Uploaded CSV ko parse karna

Complete Request Flow (Step by Step):

Step	Action	Technology
1	User CSV select karta hai Streamlit pe	Streamlit st.file_uploader
2	DELETE /clear call — purana data gone	requests.delete() → Spring Boot
3	POST /upload-csv — naya file bheja	requests.post() multipart/form-data
4	CSV parse hoti hai row by row	Apache Commons CSV
5	Har row validate aur save hoti hai	JPA repository.saveAll()
6	Response aata hai — success/fail count	ApiResponse JSON
7	Cache clear hoti hai Streamlit mein	st.cache_data.clear()
8	Analytics pages fresh data dikhate hain	GET API calls + Plotly
9	AI question poochha → Ollama jawab deta hai	Spring AI + Ollama LLM

10	PDF button → bytes generate → download	iTextPDF ByteArrayOutputStream

PART 3: TECHNOLOGIES — DETAIL MEIN

BACKEND TECHNOLOGIES (Java)

1. Spring Boot 3

Hinglish (Samjho):

Spring Boot ek Java framework hai jo REST APIs banane ke liye use hota hai. Hamare project mein Controller-Service-Repository layered architecture follow kiya.

English (Interview mein bolna):

> "Spring Boot 3 is the core backend framework. It follows layered architecture — Controller handles HTTP requests, Service contains business logic, Repository manages DB operations."

Jo padhna chahiye:

- *@RestController, @RequestMapping, @GetMapping, @PostMapping*
- *Constructor Injection (Dependency Injection)*
- *application.yml configuration*
- *REST API concepts — GET, POST, DELETE, HTTP Status Codes*

2. Spring AI + Ollama

Hinglish (Samjho):

Spring AI ek library hai jo Java apps mein LLM integrate karne ke liye use hoti hai. Ollama LLM ko locally run karta hai — koi data bahar nahi jaata!

English (Interview mein bolna):

> "Spring AI provides seamless LLM integration. Ollama runs the model locally ensuring data privacy — no business data leaves the system."

Jo padhna chahiye:

- *What is LLM (Large Language Model)?*
- *What is Ollama? (Local AI runner)*
- *Why local LLM over cloud? (Privacy)*
- *Future: RAG (Retrieval Augmented Generation)*

3. Spring Data JPA + Hibernate

Hinglish (Samjho):

JPA Java objects ko database tables se map karta hai. Humne @Entity CarSales class banai — automatic MySQL mein car_sales table ban gayi!

English (Interview mein bolna):

> "Spring Data JPA with Hibernate serves as ORM layer. Custom JPQL queries were written for complex aggregations like brand/color/state wise counts."

Jo padhna chahiye:

- *ORM concept — Object Relational Mapping*
- *@Entity, @Table, @Column, @Id, @GeneratedValue*
- *@Query annotation for custom JPQL*

• *Java records for DTOs (YearlyCountDto, MonthlyCountDto)*

4. MySQL

Hinglish (Samjho):

Relational database jahan saara data store hota hai. car_sales table mein har car ki information hai — brand, color, price, state, year etc.

English (Interview mein bolna):

> "MySQL serves as the primary relational database. Data isolation is implemented by truncating the table before each new upload."

Jo padhna chahiye:

- *Basic SQL — GROUP BY, ORDER BY, COUNT, AVG*
- *Primary Key concept*
- *Aggregate functions*

5. iTextPDF 5

Hinglish (Samjho):

Java library jo programmatically PDF files generate karne ke liye use hoti hai. "Generate PDF" click karne pe Spring Boot PDF banata hai with metrics + tables.

English (Interview mein bolna):

> "iTextPDF 5 generates PDFs dynamically. ByteArrayOutputStream stores bytes in memory, returned as ResponseEntity with Content-Disposition: attachment header."

Jo padhna chahiye:

- *Document, PdfWriter, Paragraph, PdfPTable*
- *ByteArrayOutputStream — memory mein PDF store*
- *ResponseEntity — HTTP mein bhejne ke liye*
- *Wrong import issue — javax.swing vs com.itextpdf*

FRONTEND & DATA SCIENCE (Python)

6. Streamlit

Hinglish (Samjho):

Python library jo data dashboards banati hai bina HTML/CSS ke! Hamare project mein navigation, file upload, charts, metrics sab Streamlit se hain.

English (Interview mein bolna):

> "Streamlit serves as the frontend framework enabling rapid dashboard development using pure Python. @st.cache_data optimizes API call frequency."

Jo padhna chahiye:

- *st.title, st.metric, st.dataframe, st.plotly_chart*
- *st.cache_data — caching kya hai aur kyun*
- *st.file_uploader, st.sidebar, st.spinner*
- *st.success, st.error, st.warning*

7. Plotly Express

Hinglish (Samjho):

Interactive charts banane ke liye Python library. Line chart, Bar chart, Donut Pie chart use kiye. User hover/zoom kar sakta hai.

English (Interview mein bolna):

> "Plotly Express chosen for interactivity. Charts include line/bar/donut pie using plotly_dark template matching dashboard theme."

Jo padhna chahiye:

- *px.line(), px.bar(), px.pie()*
- *update_layout() — customization*
- *template="plotly_dark"*
- *Plotly vs Matplotlib difference*

8. scikit-learn (ML)

Hinglish (Samjho):

Python ML library. Linear Regression use ki — past yearly sales se pattern seekha aur future predict kiya. R2 score = model accuracy.

English (Interview mein bolna):

> "scikit-learn LinearRegression trained on historical yearly data. Years = feature (X), sales count = target (Y). R2 score shown as accuracy."

Jo padhna chahiye:

- *What is Machine Learning? Supervised vs Unsupervised*
- *What is Linear Regression?*
- *R2 score — coefficient of determination*
- *model.fit(), model.predict(), model.score()*
- *numpy reshape(-1,1) — kyun karna padta hai*

PART 4: KEY DESIGN DECISIONS — KYU YE CHOICES LI?

Q: Kyun Spring Boot + Python dono use kiya?

Hinglish: Spring Boot enterprise REST APIs ke liye best hai. Python data science ecosystem (Plotly, sklearn) ke liye best hai. Dono combine karke best of both worlds mila.

English: "Polyglot architecture — each language used for what it does best. Java handles enterprise concerns, Python handles data science."

Q: Kyun Data Isolation implement ki?

Hinglish: Do CSV upload karne pe dono ka data mix ho gaya. AI ne 26 red cars bataye jab 13 hone chahiye the. DELETE endpoint se upload se pehle DB clear hota hai.

English: "Data isolation ensures fresh analysis per upload. A clear-before-upload mechanism prevents dataset contamination."

Q: Kyun Local LLM (Ollama)?

Hinglish: Cloud LLMs mein data bahar jaata hai — privacy concern. Ollama locally run hota hai — data secure, internet ki zaroorat nahi.

English: "Ollama ensures data privacy as no business data leaves the system — critical for sensitive dealership data."

Q: Kyun Linear Regression?

Hinglish: Simple, interpretable model. Kam data mein bhi kaam karta hai. Explainable AI. Future: ARIMA/Prophet for better time-series.

English: "Linear Regression chosen for interpretability. Future scope includes ARIMA or Facebook Prophet for seasonal forecasting."

PART 5: REAL PROBLEMS SOLVED

Problem	Solution	Feature
Manual report banana	Auto PDF generation	PDF Reports
Data mixing in DB	Clear before upload	Data Isolation
Complex SQL queries	Natural language AI	AI Insights
No future planning	ML forecast	Sales Prediction
Multi-dim analysis hard	Brand/Color/State charts	Deep Analytics
KPI scatterred	Single KPI page	KPI Summary

QUICK REVISION — TECH STACK

Backend (Java)	Spring Boot 3 | Spring AI | Ollama | MySQL | JPA | iTextPDF | Commons CSV
Frontend (Python)	Streamlit | Plotly Express | scikit-learn | Pandas | NumPy | Requests
Architecture	Controller → Service → Repository (Layered)
API Style	REST — GET/POST/DELETE with JSON responses
DB	MySQL — car_sales table, JPQL custom queries
AI	Spring AI + Ollama (local LLM, no internet needed)
ML	Linear Regression — years as X, sales as Y, R2 score
PDF	iTextPDF → ByteArrayOutputStream → ResponseEntity<byte[]>